```
/* Over the primed indices */
$X^{\tau'k'}_{ilj} = 0$
$Z^{\tau'k'}_{m'_x n'_x lj} = 0$
forall $n'_x$ do
  forall $m'_x$ do
    $Y^{\tau'k'}_{n'_z m'_z j} = 0$
    forall $n'_z$ do
      forall $m'_z$ do
        $D^{\tau'k'}_{n'_x m'_x n'_z m'_z n'_y m'_y} = 0$
        forall $n'_y$ do
          forall $m'_y$ do
            $D^{\tau'k'}_{n'_x m'_x n'_z m'_z n'_y m'_y}$
        forall $n'_y$ do
          forall $m'_y \neq \{n_{y'}...\min(N - n'_x - n'_z, N - m'_x - m'_z)\}$ do
            $D^{\tau'k'}_{n'_x m'_x n'_z m'_z (n'_y m'_y)} = \frac{1}{2}(D^{\tau'k'}_{n'_x m'_x n'_z m'_z n'_y m'_y} + D^{\tau'k'}_{n'_x m'_x n'_z m'_z m'_y n'_y})$
            $D^{\tau'k'}_{n'_x m'_x n'_z m'_z [n'_y m'_y]} = \frac{1}{2}(D^{\tau'k'}_{n'_x m'_x n'_z m'_z n'_y m'_y} - D^{\tau'k'}_{n'_x m'_x n'_z m'_z m'_y n'_y})$
            forall $j_+$ do
              $Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_+} = \sum_{m'_y n'_y} D^{\tau'k'}_{n'_x m'_x n'_z m'_z (n'_y m'_y)} A^y_{(m'_y n'_y), j_+} \lambda_{j_+}$
            forall $j_-$ do
              $Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_-} = \sum_{m'_y n'_y} D^{\tau'k'}_{n'_x m'_x n'_z m'_z [n'_y m'_y]} A^y_{[m'_y n'_y], j_-} \lambda_{j_-}$
    forall $j_+$ do
      forall $n'_z$ do
        forall $m'_z \neq \{n_{z'}...\min(N - n'_x, N - m'_x)\}$ do
          $Y^{\tau'k'}_{n'_x m'_x (n'_z m'_z) j_+} = \frac{1}{2}(Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_+} + Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_+})$
          $Y^{\tau'k'}_{n'_x m'_x [n'_z m'_z] j_+} = \frac{1}{2}(Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_+} - Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_+})$
          forall $l_+$ do
            $Z^{\tau'k'}_{n'_x m'_x l_+ j_+} = \sum_{m'_z n'_z} Y^{\tau'k'}_{n'_x m'_x (n'_z m'_z) j_+} A^z_{(m'_z n'_z), l_+} \lambda_{l_+}$
          forall $l_-$ do
            $Z^{\tau'k'}_{n'_x m'_x l_- j_+} = \sum_{m'_z n'_z} Y^{\tau'k'}_{n'_x m'_x [n'_z m'_z] j_+} A^z_{[m'_z n'_z], l_-} \lambda_{l_-}$
    forall $j_-$ do
      forall $n'_z$ do
        forall $m'_z \neq \{n_{z'}...\min(N - n'_x, N - m'_x)\}$ do
          $Y^{\tau'k'}_{n'_x m'_x (n'_z m'_z) j_-} = \frac{1}{2}(Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_-} + Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_-})$
          $Y^{\tau'k'}_{n'_x m'_x [n'_z m'_z] j_-} = \frac{1}{2}(Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_-} - Y^{\tau'k'}_{n'_x m'_x n'_z m'_z j_-})$
          forall $l_+$ do
            $Z^{\tau'k'}_{n'_x m'_x l_+ j_-} = \sum_{m'_z n'_z} Y^{\tau'k'}_{n'_x m'_x (n'_z m'_z) j_-} A^z_{(m'_z n'_z), l_+} \lambda_{l_+}$
          forall $l_-$ do
            $Z^{\tau'k'}_{n'_x m'_x l_- j_-} = \sum_{m'_z n'_z} Y^{\tau'k'}_{n'_x m'_x [n'_z m'_z] j_-} A^z_{[m'_z n'_z], l_-} \lambda_{l_-}$
forall $n'_x$ do
  forall $m'_x \leq n'_x$ do
    forall $j_+$ do
      forall $l_+$ do
        $Z_{(n'_x m'_x) l_+ j_+} = \frac{1}{2}(Z_{n'_x m'_x l_+ j_+} + Z_{m'_x n'_x l_+ j_+})$
        $Z_{[n'_x m'_x] l_+ j_+} = \frac{1}{2}(Z_{n'_x m'_x l_+ j_+} - Z_{m'_x n'_x l_+ j_+})$
        forall $i_+$ do
          $X^{\tau'k'}_{i_+ l_+ j_+} = \sum_{m'_x n'_x} Z^{\tau'k'}_{(n'_x m'_x) l_+ j_+} A^x_{(m'_x n'_x), i_+} \lambda_{i_+}$
        forall $i_-$ do
          $X^{\tau'k'}_{i_- l_+ j_+} = \sum_{m'_x n'_x} Z^{\tau'k'}_{[n'_x m'_x] l_+ j_+} A^x_{[m'_x n'_x], i_-} \lambda_{i_-}$
      forall $l_-$ do
        $Z_{(n'_x m'_x) l_- j_+} = \frac{1}{2}(Z_{n'_x m'_x l_- j_+} + Z_{m'_x n'_x l_- j_+})$
        $Z_{[n'_x m'_x] l_- j_+} = \frac{1}{2}(Z_{n'_x m'_x l_- j_+} - Z_{m'_x n'_x l_- j_+})$
        forall $i_+$ do
          $X^{\tau'k'}_{i_+ l_- j_+} = \sum_{m'_x n'_x} Z^{\tau'k'}_{(n'_x m'_x) l_- j_+} A^x_{(m'_x n'_x), i_+} \lambda_{i_+}$
        forall $i_-$ do
          $X^{\tau'k'}_{i_- l_- j_+} = \sum_{m'_x n'_x} Z^{\tau'k'}_{[n'_x m'_x] l_- j_+} A^x_{[m'_x n'_x], i_-} \lambda_{i_-}$
    forall $j_-$ do
      forall $l_+$ do
        $Z_{(n'_x m'_x) l_+ j_-} = \frac{1}{2}(Z_{n'_x m'_x l_+ j_-} + Z_{m'_x n'_x l_+ j_-})$
        $Z_{[n'_x m'_x] l_+ j_-} = \frac{1}{2}(Z_{n'_x m'_x l_+ j_-} - Z_{m'_x n'_x l_+ j_-})$
        forall $i_+$ do
          $X^{\tau'k'}_{i_+ l_+ j_-} = \sum_{m'_x n'_x} Z^{\tau'k'}_{(n'_x m'_x) l_+ j_-} A^x_{(m'_x n'_x), i_+} \lambda_{i_+}$
        forall $i_-$ do
          $X^{\tau'k'}_{i_- l_+ j_-} = \sum_{m'_x n'_x} Z^{\tau'k'}_{[n'_x m'_x] l_+ j_-} A^x_{[m'_x n'_x], i_-} \lambda_{i_-}$
      forall $l_-$ do
        $Z_{(n'_x m'_x) l_- j_-} = \frac{1}{2}(Z_{n'_x m'_x l_- j_-} + Z_{m'_x n'_x l_- j_-})$
        $Z_{[n'_x m'_x] l_- j_-} = \frac{1}{2}(Z_{n'_x m'_x l_- j_-} - Z_{m'_x n'_x l_- j_-})$
        forall $i_+$ do
          $X^{\tau'k'}_{i_+ l_- j_-} = \sum_{m'_x n'_x} Z^{\tau'k'}_{(n'_x m'_x) l_- j_-} A^x_{(m'_x n'_x), i_+} \lambda_{i_+}$
        forall $i_-$ do
          $X^{\tau'k'}_{i_- l_- j_-} = \sum_{m'_x n'_x} Z^{\tau'k'}_{[n'_x m'_x] l_- j_-} A^x_{[m'_x n'_x], i_-} \lambda_{i_-}$
/* Over the unprimed indices */
$\tilde{X}^{\tau'k'}_{m_x n_x j l} = 0$
forall $n_x$ do
  forall $m_x \leq n'_x$ do
    forall $j_+$ do
      forall $l_+$ do
        $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = 0$
        $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = 0$
        forall $i_+$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = \sum_{i_+} X^{\tau'k'}_{i_+ l_+ j_+} A^x_{i_+, m_x n_x}$
        forall $i_-$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = \sum_{i_-} X^{\tau'k'}_{i_- l_+ j_+} A^x_{i_-, m_x n_x}$
        $\tilde{X}^{\tau'k'}_{m_x n_x j_+ l_+} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} + \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
        $\tilde{X}^{\tau'k'}_{n_x m_x j_+ l_+} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} - \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
      forall $l_-$ do
        $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = 0$
        $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = 0$
        forall $i_+$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = \sum_{i_+} X^{\tau'k'}_{i_+ l_- j_+} A^x_{i_+, m_x n_x}$
        forall $i_-$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = \sum_{i_-} X^{\tau'k'}_{i_- l_- j_+} A^x_{i_-, m_x n_x}$
        $\tilde{X}^{\tau'k'}_{m_x n_x j_+ l_-} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} + \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
        $\tilde{X}^{\tau'k'}_{n_x m_x j_+ l_-} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} - \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
    forall $j_-$ do
      forall $l_+$ do
        $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = 0$
        $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = 0$
        forall $i_+$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = \sum_{i_+} X^{\tau'k'}_{i_+ l_+ j_-} A^x_{i_+, m_x n_x}$
        forall $i_-$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = \sum_{i_-} X^{\tau'k'}_{i_- l_+ j_-} A^x_{i_-, m_x n_x}$
        $\tilde{X}^{\tau'k'}_{m_x n_x j_- l_+} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} + \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
        $\tilde{X}^{\tau'k'}_{n_x m_x j_- l_+} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} - \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
      forall $l_-$ do
        $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = 0$
        $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = 0$
        forall $i_+$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{sym}} = \sum_{i_+} X^{\tau'k'}_{i_+ l_- j_-} A^x_{i_+, m_x n_x}$
        forall $i_-$ do
          $\tilde{X}^{\tau'k'}_{\mathrm{antisym}} = \sum_{i_-} X^{\tau'k'}_{i_- l_- j_-} A^x_{i_-, m_x n_x}$
        $\tilde{X}^{\tau'k'}_{m_x n_x j_- l_-} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} + \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
        $\tilde{X}^{\tau'k'}_{n_x m_x j_- l_-} = \tilde{X}^{\tau'k'}_{\mathrm{sym}} - \tilde{X}^{\tau'k'}_{\mathrm{antisym}}$
forall $n_x$ do
  forall $m_x$ do
    $\tilde{Z}^{\tau'k'}_{j m_z n_z} = 0$
    forall $j_+$ do
      forall $n_z$ do
        forall $m_z \neq \{n_z...\min(N - n_x, N - m_x\}$ do
          $\tilde{Z}^{\tau'k'}_{\mathrm{sym}} = 0$
          $\tilde{Z}^{\tau'k'}_{\mathrm{antisym}} = 0$
          forall $l_+$ do
            $\tilde{Z}^{\tau'k'}_{\mathrm{sym}} = \sum_{l_+} \tilde{X}^{\tau'k'}_{m_x n_x j_+ l_+} A^z_{l_+, m_z n_z}$
          forall $l_-$ do
            $\tilde{Z}^{\tau'k'}_{\mathrm{antisym}} = \sum_{l_-} \tilde{X}^{\tau'k'}_{m_x n_x j_- l_-} A^z_{l_-, m_z n_z}$
          $\tilde{Z}^{\tau'k'}_{m_x n_x j_+ m_z n_z} = \tilde{Z}^{\tau'k'}_{\mathrm{sym}} + \tilde{Z}^{\tau'k'}_{\mathrm{antisym}}$
          if $m_z < n_z$ then
            $\tilde{Z}^{\tau'k'}_{n_x m_x j_+ m_z n_z} = \tilde{Z}^{\tau'k'}_{\mathrm{sym}} - \tilde{Z}^{\tau'k'}_{\mathrm{antisym}}$
    forall $n_z$ do
      forall $m_z \neq \{n_z...\min(N - n_x, N - m_x)\}$ do
        $\tilde{Z}^{\tau'k'}_{\mathrm{sym}} = 0$
        $\tilde{Z}^{\tau'k'}_{\mathrm{antisym}} = 0$
        forall $l_+$ do
          $\tilde{Z}^{\tau'k'}_{\mathrm{sym}} = \sum_{l_+} \tilde{X}^{\tau'k'}_{m_x n_x j_- l_+} A^z_{l_+, m_z n_z}$
        forall $l_-$ do
          $\tilde{Z}^{\tau'k'}_{\mathrm{antisym}} = \sum_{l_-} \tilde{X}^{\tau'k'}_{m_x n_x j_- l_-} A^z_{l_-, m_z n_z}$
        $\tilde{Z}^{\tau'k'}_{m_x n_x j_+ m_z n_z} = \tilde{Z}^{\tau'k'}_{\mathrm{sym}} + \tilde{Z}^{\tau'k'}_{\mathrm{antisym}}$
        if $m_z < n_z$ then
          $\tilde{Z}^{\tau'k'}_{n_x m_x j_+ m_z n_z} = \tilde{Z}^{\tau'k'}_{\mathrm{sym}} - \tilde{Z}^{\tau'k'}_{\mathrm{antisym}}$
    forall $n_z$ do
      forall $m_z$ do
        forall $n_y$ do
          forall $m_y \neq \{n_y...\min(N - n_x - n_z, N - m_x - m_z)\}$ do
            $\tilde{Y}^{\tau'k'}_{\mathrm{sym}} = 0$
            $\tilde{Y}^{\tau'k'}_{\mathrm{antisym}} = 0$
            forall $j+$ do
              $\tilde{Y}^{\tau'k'}_{\mathrm{sym}} = \sum_{j_+} \tilde{Z}^{\tau'k'}_{m_x n_x j_+ m_z n_z} A^y_{j_+, m_y n_y}$
            forall $j_-$ do
              $\tilde{Y}^{\tau'k'}_{\mathrm{antisym}} = \sum_{j_-} \tilde{Z}^{\tau'k'}_{m_x n_x j_- m_z n_z} A^y_{j_-, m_y n_y}$
            $\tilde{Y}^{\tau'k'}_{m_y n_y m_z n_z m_x n_x} = \tilde{Y}^{\tau'k'}_{\mathrm{sym}} + \tilde{Y}^{\tau'k'}_{\mathrm{antisym}}$
            if $m_y < n_y$ then
              $\tilde{Y}^{\tau'k'}_{n_y m_y m_z n_z m_x n_x} = \tilde{Y}^{\tau'k'}_{\mathrm{sym}} - \tilde{Y}^{\tau'k'}_{\mathrm{antisym}}$
        forall $n_y$ do
          forall $m'_y$ do
            $E^{\tau'k'\tau k}_{\mathrm{dir}} = \sum_{\boldsymbol{m},\boldsymbol{n}} D^{\tau k}_{m_y n_y m_z n_z m_x n_x} \tilde{Y}^{\tau'k'}_{m_y n_y m_z n_z m_x n_x}$
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